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Cart263

# Project Atermis II

The Artemis II website was designed to show the path taken by the rocket for the lunar mission to make it easier to understand. Instead of showing only text and images, the website uses a 3D scene where people can watch the rocket travel from Earth to the Moon and back. The final project includes Earth, the Moon, stars, and a launch button that starts the animation. As the rocket moves, the camera follows it smoothly and mission information updates on screen. This makes the website feel active and immersive. The project shows how web design can be used to explain scientific ideas in a more visual and interesting format. Overall, it became both an educational tool and a creative digital experience.

Three.js was the main library used to build the website because it makes creating 3D graphics in the browser much easier. For a website following a curved trajectory, using 3D elements makes it much more enjoyable to watch. In this project, Three.js was used to create Earth, the Moon, the rocket, lights, and camera movement. Basic shapes such as spheres, cylinders, and cones were combined to build the rocket model. Textures were added to make the planets look realistic, only the lday part of the earth to keep it separate from the background. Lighting helped give depth to the scene, while the renderer displayed everything inside the HTML canvas.

Animation was created using requestAnimationFrame, which keeps movement smooth. The rocket follows a curved path using CatmullRomCurve3, which was useful for recreating the real mission route. Camera lerping was used so the camera follows the rocket smoothly instead of jumping suddenly. Textures were applied to Earth and the Moon to improve realism. JavaScript DOM updates were also used to change the mission phase and distance text live on screen. A random star field was created in the

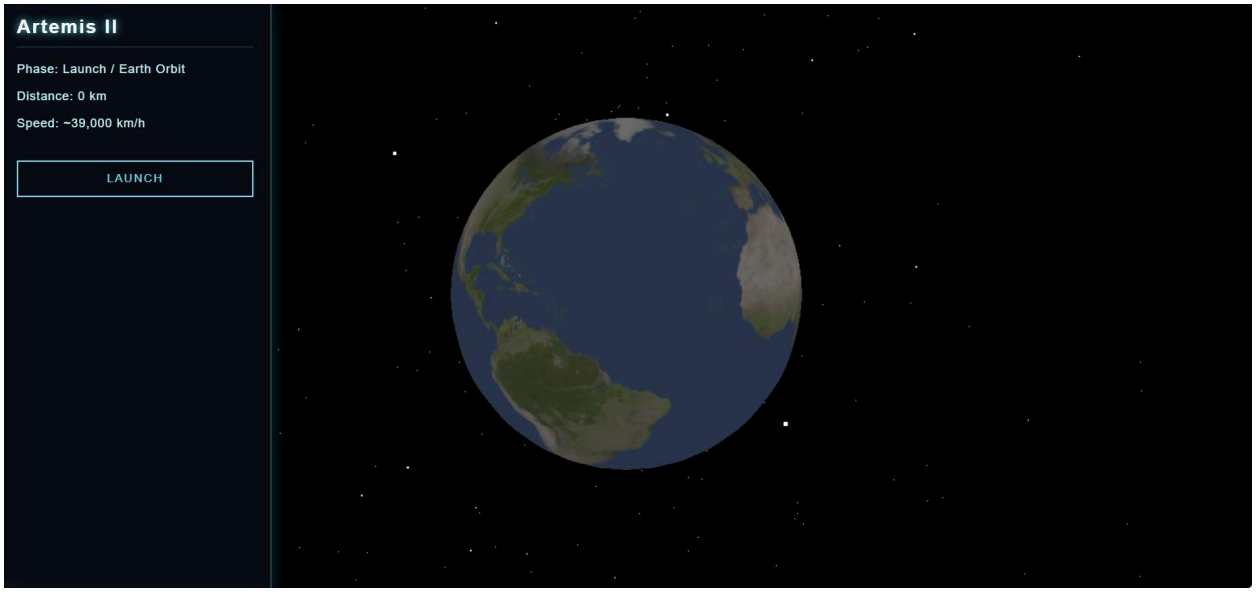
background to give the feeling of space. These techniques helped make the website feel interesting and dynamic.

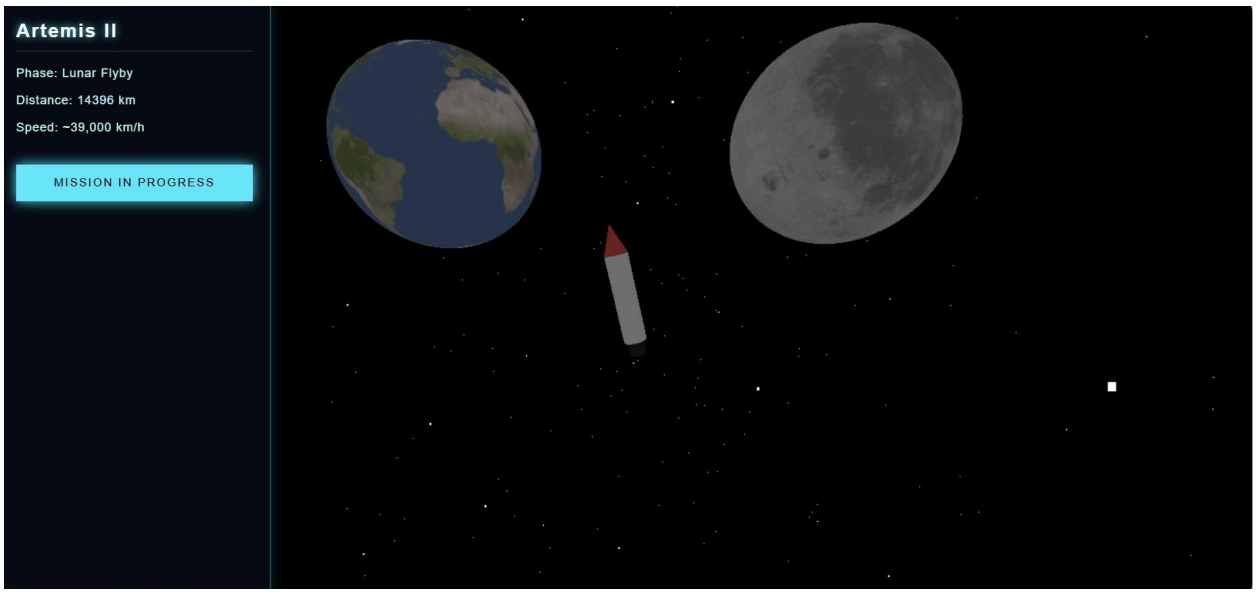
The website was built using HTML, CSS, JavaScript, and Three.js. First, the scene, camera, and renderer were created. Then Earth and the Moon were added using sphere geometry and textures. The rocket was built from simple shapes grouped together so it could move as one object. A custom curve path was then created for the mission route. When the user presses the launch button, the rocket begins moving along this path. During the animation, the rocket rotates to face the correct direction, while the camera follows behind it. Text updates on screen show the mission phase and distance traveled. Resize settings were also added so the project works on different screen sizes as an added bonus.

The main intention of the project was to help people understand the Artemis II mission in a more visual and accessible way. Many people do not fully understand how spacecraft travel, so the website turns the mission into something easy to follow. People can watch the rocket leave Earth, fly around the Moon, and return home. This helps explain that space travel uses curved orbital paths rather than straight lines. The project also highlights the meaning of Artemis II as the first crewed mission around the Moon since Apollo. It represents progress, exploration, and preparation for future missions to the Moon and Mars.

The inspiration for the project was, of course, the recent lunar mission that took place in April. According to the Agence spatiale canadienne, why this mission is important comes back to the fact that it was the first crewed lunar mission since 1972; and it would be the furthest humans had ever been in space, 406,771 km from earth and with a multitude of firsts. First person of color, Victor Glover, first woman, Christina Koch, and first non US citizen, Jeremy Hansen from the Canadian Space Agency, to travel beyond low earth orbit and around the moon, led by commander Reid Wiseman.
Another piece of media that inspired me to make my project space themed was the book I was reading at the time, Project Hail Mary by Andy Weir, an author I've been a fan of for a few years. The release of the movie based on the book was happening

around the same time as the start of the mission which seemed to help both events in rising to popularity. I absolutely adored the book and when I finally got to see the movie, the Artemis 2 mission was on its way back to earth. This definitely helped me understand some of the harder concepts regarding the mission and furthered my interest in anything space related. The book is about someone sent on a mission so far from earth that there is no hope to come back in order to save humanity and not only is the science of it all well explained and a funny read, it talks about grief and sacrifice, the immensity of space, the length people will go for to help each other and the connection made. Truly inspiring to see people right now experiencing that with the added bonus of making it back to earth.


Artemis II
Phase: Launch / Earth Orbit
Distance: 0 km
Speed: ~39,000 km/h
LAUNCH


Artemis II
Phase: Lunar Flyby
Distance: 14396 km
Speed: ~39,000 km/h
MISSION IN PROGRESS

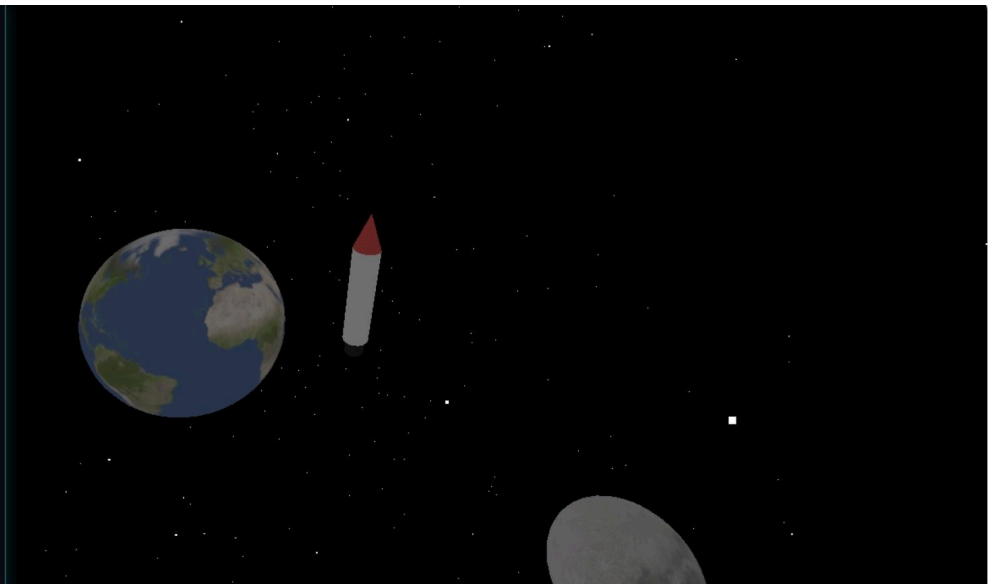
## Artemis II

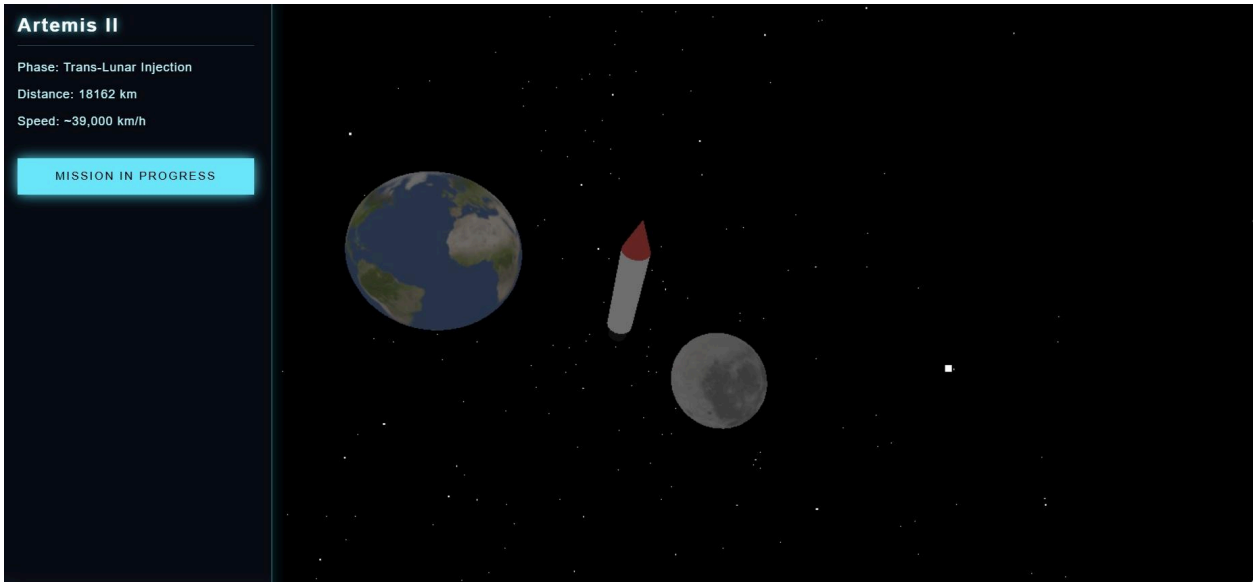
Phase: Launch / Earth Orbit

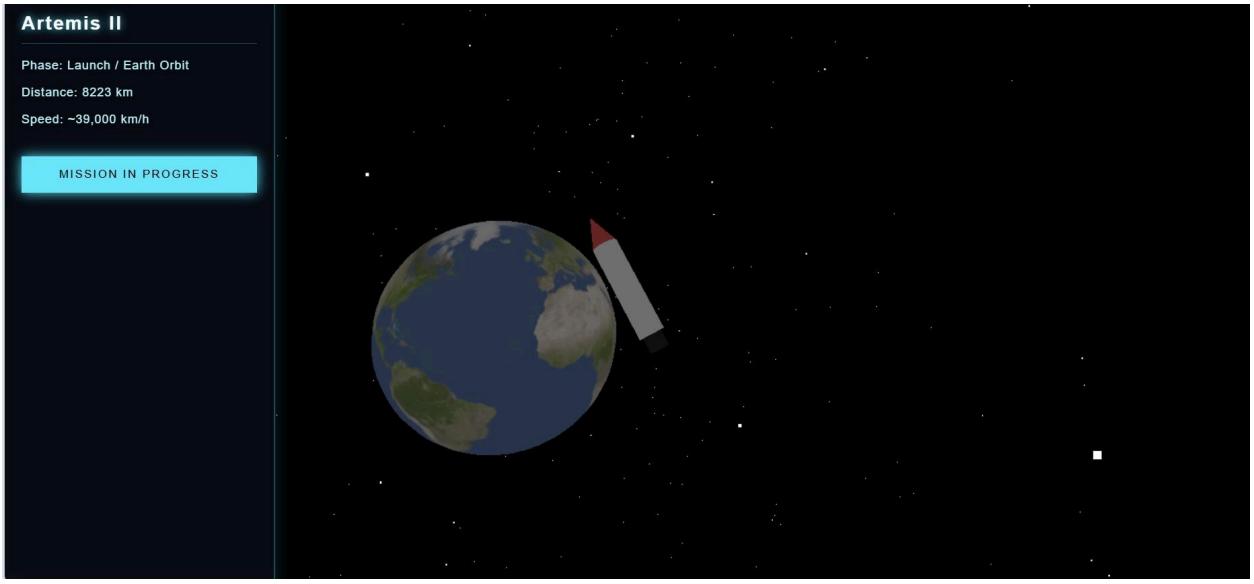
Distance: 13503 km

Speed: ~39,000 km/h

MISSION IN PROGRESS




Artemis II
Phase: Trans-Lunar Injection
Distance: 18162 km
Speed: ~39,000 km/h
MISSION IN PROGRESS


Artemis II
Phase: Launch / Earth Orbit
Distance: 8223 km
Speed: ~39,000 km/h
MISSION IN PROGRESS